

CHE 205 HW 6

Due: 11.59 pm on 04/05/2020

Exercise 5.9: Penicillin Fermentation

A model for a batch reactor in which penicillin is produced by fermentation has been derived as follows (Constantinides et al. 1970) for cell production and penicillin synthesis, respectively:

$$\begin{aligned}\frac{dy_1}{dt} &= b_1 y_1 - \frac{b_1}{b_2} y_1^2 & y_1(0) &= 0.03 \\ \frac{dy_2}{dt} &= b_3 y_1 & y_2(0) &= 0.0\end{aligned}\tag{5.46}$$

where

y_1 = dimensionless concentration of cell mass

y_2 = dimensionless concentration of penicillin

t = dimensionless time, $0 \leq t \leq 1$

Experiments have determined that

$$b_1 = 13.1$$

$$b_2 = 0.94$$

$$b_3 = 1.71$$

- a. Solve this ODE-IVP using the Euler method and Excel (not VBA).
- b. Solve this ODE-IVP using the second-order RK method and Excel (not VBA).
- c. Solve this ODE-IVP using the Euler VBA program as given in Example 5.7.
- d. Solve this ODE-IVP using the second-order RK VBA program from Exercise 5.8.

In all cases, experiment with the time step to assure an accurate solution. Also, graph y_1 and y_2 versus time with appropriate annotations.

e. Solve this ODE-IVO using fourth-order RK method and Excel (not VBA)

Each subsection is 20 points.